Five explicit Higman–Sims doublings and three source-link qualifications

L091 finite construction component

We use the archived supplement dated 6 October 2003 to D. Leemans [1], with its accepted coordinates in the 86-class literal coset catalogue. All labels below are supplement labels. The main paper calls the construction of [2, Theorem 4.1] doubling. We use precisely the two-point-shadow operation in the acquired theorem adapter: the prior primary reading included that theorem and its construction, but did not obtain the complete article. No different convention is silently imposed. The elementary argument needed here is given in full.

Theorem 1. *The following five printed links have explicit incidence isomorphisms from the indicated doubled parents to the target geometries:*

$$3.2 \longrightarrow 3.3, \quad 3.4 \longrightarrow 3.5, \quad 3.8 \longrightarrow 3.9, \quad 4.8 \longrightarrow 4.10, \quad 4.14 \longrightarrow 4.19.$$

The other three printed links, $3.12 \to 3.15$*,* $3.13 \to 3.16$ *and* $3.20 \to 3.21$*, are not applications of this operation to the stated parents. The first two fail its two-point hypothesis at the only type choice compatible with the target sizes; the third fails the necessary type-size condition for every choice.*

These are ancillary construction qualifications, relative to the explicit operation below. They neither refute the principal classification nor claim that the three target geometries do not exist. Their existence and coordinates remain acquired. No repair using an unspecified alternative doubling convention or an unverified replacement parent is asserted.

1 The construction and exact finite test

Lemma 1. *Suppose a flag-transitive geometry has types* X, E *such that the* X*-shadows of its* E*-objects are distinct two-element sets. Suppose, for any other object* z*,*

$$e * z \quad \Longleftrightarrow \quad \sigma_X(e) \subseteq \sigma_X(z). \tag{1}$$

Replace X, E *by two copies* X^+, X^- *of* X*. Join* x^+ *to* y^- *exactly when* $\{x, y\}$ *is an* E*-shadow; join either copy of* x *to* z *exactly when* $x * z$ *originally. Leave other incidences unchanged. This is a geometry with the same chamber action as the original one.*

Proof. An old chamber (x, e, F) determines the new chamber (x^+, y^-, F), where y is the other endpoint of e. Conversely a new flag using both copies determines its unique edge, and (1) makes that edge incident with all its unchanged objects. It therefore extends from an old flag to an old chamber and then to a new chamber. A flag using only one copy extends from the corresponding old point flag; if neither copy occurs, extend its unchanged flag first. This proves chamber extension in every case and the asserted equivariant chamber bijection. Chamber transitivity then implies flag transitivity. The argument also works in rank two; it does not assert extra rank-two connectivity hypotheses from a theorem stated only in higher rank. □

Write $G = \mathrm{HS}$. Objects are right cosets; incidence means nonempty intersection. We also denote the base point and edge stabilizers by X, E. The number of points incident with E is

$[E : E \cap X]$. When this is two, choose a recorded $h \in E \setminus X$; the endpoints are X and Xh. Their pointwise stabilizer is $X \cap X^h$, where $X^h = h^{-1}Xh$. Hence distinct edge shadows are equivalent to

$$2|X \cap X^h| = |E|. \tag{2}$$

Indeed, E is contained in the pair's setwise stabilizer and interchanges its endpoints. That setwise stabilizer has twice the order of the pointwise stabilizer, so (2) gives equality. The resulting map from E-cosets to pairs is therefore injective.

For any remaining base stabilizer Z, take right transversals T_X of $X \cap Z$ in X and T_E of $E \cap Z$ in E. The following finite conditions are equivalent to (1):

$$ha^{-1} \in XZ \iff a^{-1} \in EZ \qquad (a \in T_X), \tag{3}$$

$$a^{-1} \in XZ \text{ and } ha^{-1} \in XZ \qquad (a \in T_E). \tag{4}$$

Here XZ and EZ are identity double cosets, not asserted subgroups. The objects Za, $a \in T_X$, are exactly the Z-objects incident with X. The second endpoint meets Za precisely when $ha^{-1} \in XZ$, while E meets it precisely when $a^{-1} \in EZ$. This proves sufficiency of a pair of endpoints in (3). Equation (4) checks necessity for every Z-neighbor of E, including those not assumed to meet X. Transitivity transports these tests to all edges. No enumeration of the 44 352 000 ambient elements is needed.

2 Five literal target maps

The new geometry has base stabilizers

$$X,\ X^h,\ (Z_i)_{i \neq X,E}. \tag{5}$$

In the minus copy the base point is Xh, so the right coset $X^h g$ maps to Xhg. The chamber argument makes every type-pair incidence a single G-orbit, and thus (5) is its full coset description. Each certificate supplies an actual n in the acquired normalizer N of G and a type permutation, and checks that conjugating every subgroup in (5) by n gives the target family in that order. Consequently $Hg \mapsto H^n g^n$ is a well-defined typewise bijection preserving incidence and nonincidence. This is an exact incidence map, not an inference from a diagram or subgroup order.

The following table uses internal catalogue type indices solely to make the literal certificates easy to locate. The source-coordinate ledger also supplies the corresponding native node indices.

Parent→target	Internal parent→target	(X, E)	(3) tests	(4) tests
$3.2 \to 3.3$	$35 \to 34$	$(3, 1)$	8	2
$3.4 \to 3.5$	$32 \to 33$	$(3, 2)$	15	3
$3.8 \to 3.9$	$30 \to 28$	$(3, 1)$	8	2
$4.8 \to 4.10$	$70 \to 69$	$(4, 3)$	2310	48
$4.14 \to 4.19$	$54 \to 57$	$(2, 3)$	2800	55

The two rank-four rows test both unchanged types. The accepted target geometries transport all their residue properties through these maps. The present component does not infer full automorphism or correlation groups merely from the induced specified G-action.

3 Three exact obstructions and evidence scope

For the links $3.12 \to 3.15$ and $3.13 \to 3.16$, the parent and target size lists, in the same order, are

$$(352, 30800, 1100) \longrightarrow (352, 1100, 1100),$$
$$(352, 30800, 5600) \longrightarrow (352, 5600, 5600).$$

Thus the only possible choices duplicate the 1100- or 5600-object type and remove the 30800-object type. In each case the edge stabilizer has order 1440, and its intersection with the proposed point stabilizer has order 144. Each edge shadow therefore has ten points, not two. These are literal subgroup intersections, also acquired in the numerical metadata. In particular a type permutation cannot fix the obstruction.

For $3.20 \to 3.21$, the parent and target size lists are

$$(15400, 346500, 15400) \quad \text{and} \quad (30800, 15400, 15400).$$

The operation replaces two sizes by two copies of one existing size and retains the third. It cannot create the new size 30800. The native statements are on supplement pages6,7 and8; the eight links and their source-node maps are explicitly retained.

The checker reads the exact previously accepted group, complete family and source-coordinate inputs. GAP's finite subgroup, transversal and permutation computations remain explicit computational premises. It verifies all five saved subgroup maps directly, all local tests (2)–(4), and all three obstructions. Thirty damaged certificates are rejected. An independent literal-coset control over S_3, D_8, A_4 compares the criterion with all actual pair shadows and all relevant edge–object incidences, including both successful and failed endpoint conditions. The failed first probe is preserved: it returned three positive cases before stopping at the first ten-point shadow. The changed probe records all eight outcomes; default replay performs certificate checks, not a new classification or transporter search. Source transcription, prior classifications and their retained subgroup-data premises are acquired without additional census credit.

References

[1] D. Leemans, The residually weakly primitive geometries of HS, Australas. J. Combin. **33** (2005), 231–236, and the retained 23-page supplement dated 6 October 2003.

[2] C. Lefèvre-Percsy, N. Percsy and D. Leemans, New geometries for finite groups and polytopes, Bull. Belg. Math. Soc. Simon Stevin **7** (2000), 583–610.